Lesson 2.7 Guided Notes

Session 1 · Dividing a Decimal by a Whole Number

I can divide a decimal by a whole number and place the decimal point correctly. · 6.NS.B.3

Word / Spanish	Meaning — complete the blank	Example
dividend / dividendo	The number being _____; the amount you share.	In 84.6 ÷ 6, it is 84.6
divisor / divisor	The number you _____ by.	In 84.6 ÷ 6, it is 6
quotient / cociente	The _____ to a division problem.	84.6 ÷ 6 = 14.1
annex a zero / agregar un cero	Write a 0 at the end of a decimal; the _____ stays the same.	7.2 = 7.20

Key idea

Put the decimal point in the quotient straight _____ the point in the dividend.

If an amount is left over, _____ a zero and keep dividing.

Check: quotient × divisor = _____.

Worked example

A rope 25.2 meters long is cut into 6 equal pieces. How long is each piece?

Write 25.2 ÷ 6 and put the point in the quotient straight _____ the point in 25.2.

6 goes into 25 _____ times with 1 left over. Bring down the 2 to make 12.

12 ÷ 6 = _____, so the quotient is 4.2.

Answer: Each piece is _____ m. Check: 4.2 × 6 = 25.2

You try

1. Find each quotient.
a. 8.4 ÷ 4 = _____
b. 13.5 ÷ 5 = _____
c. 7.5 ÷ 2 = _____

_____

2. Four friends share a $9.60 bill equally. How much does each pay?

_____

Sum it up: To divide a decimal by a whole number, I place the decimal point ___ because ___.

_____

Session 2 · Dividing by a Decimal

I can divide by a decimal by making the divisor a whole number first. · 6.NS.B.3

Word / Spanish	Meaning — complete the blank	Example
divisor / divisor	The number you _____ by.	In 8.4 ÷ 0.2, it is 0.2
equivalent division / división equivalente	A new division problem with the _____ quotient.	8.4 ÷ 0.2 = 84 ÷ 2
power of ten / potencia de diez	10, 100, 1,000 and so on; each one is 10 times the _____.	0.25 × 100 = 25

Key idea

Multiply the divisor and the dividend by the _____ power of 10 until the divisor is a _____ number.

The quotient does not _____: 5 ÷ 0.25 = 500 ÷ 25 = 20.

Worked example

How many 0.7-liter bottles can be filled from 8.4 liters?

Write 8.4 ÷ 0.7. The divisor has _____ decimal place.

Multiply both by 10: 0.7 becomes _____ and 8.4 becomes _____.

84 ÷ 7 = 12. Check: 12 × 0.7 = 8.4

Answer: _____ bottles

You try

1. Rewrite with a whole-number divisor. Then divide.
a. 6.4 ÷ 0.8 = _____
b. 2.7 ÷ 0.3 = _____
c. 1.8 ÷ 0.06 = _____

_____

2. How many 0.4-liter cups can be filled from 9.6 liters of juice?

_____

Sum it up: To divide by a decimal, I multiply both numbers by ___ because ___.

_____

Session 1 · Model, reason, and explain

Dividing a Decimal by a Whole Number · 6.NS.B.3

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Share 7.2 meters of ribbon equally among 3 groups. Represent 7.2 as 72 tenths, divide, then connect that model to 7.2 ÷ 3. Estimate the answer and check by multiplication.

Total: 72 tenths



_____

_____

_____

Extend your reasoning

Explain why an answer of 24 meters is unreasonable before calculating exactly.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Dividing by a Decimal · 6.NS.B.3

Success: represent the quantities, justify the relationship, and check the result with the correct units.

How many 0.6-liter bottles can be filled from 4.8 liters? Rename both measures in tenths of a liter. Explain why scaling only the divisor would change the question.

Liters	Tenths of a liter
4.8	48
0.6	6

_____

_____

_____

Extend your reasoning

Why is the quotient 8 larger than the numerical value 4.8?

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?